Trace-Triggered Speculative Councils: Protocol v0

A registered design for confidence as a control signal

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Prospective design; no live result. This document registers a proposed comparison and an executable harness. It does not claim that the trace trigger, the named model order, or cancellation economics work. The frozen MiniMax confidence study is unchanged and continues under its own budget.

Research question

For non-trivial tasks with an independently checkable completion condition, can a live trace-risk policy launch backup models early enough to preserve the quality of an always-on model council while reducing latency and billed work?

This is **speculative execution**, not ordinary answer ensembling. A model council normally obtains several completed answers and ranks, votes, or fuses them. The proposed controller begins with one observable primary, launches challengers only after persistent evidence of trouble, accepts the first candidate that passes an external verifier, and requests cancellation of the losers. Council synthesis is an exception path after the race produces no acceptable candidate.

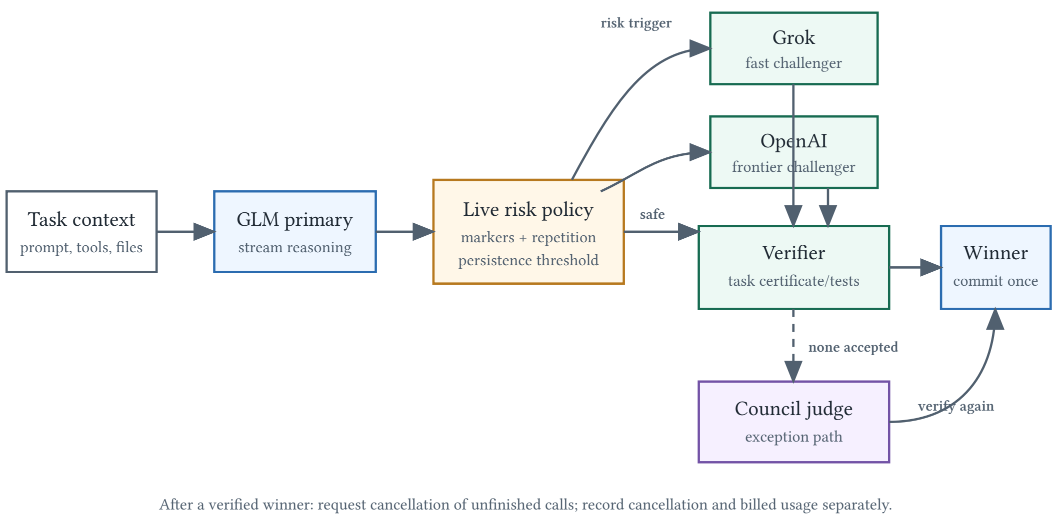

Figure 1: Trace-triggered speculative execution.

System under test

- **Primary:** an OpenAI-compatible GLM endpoint with streamed reasoning.
- **Challengers:** a fast Grok endpoint and an OpenAI frontier endpoint.

- **Handoff:** the original task, prior messages, tool results, files read, workspace revision, and policy metadata. Hidden chain of thought is not copied between providers.
- **Side effects:** paused before challenger launch. Candidates work in read-only lanes or isolated worktrees. Only an elected candidate may commit external changes.
- **Verification:** task-specific tests, certificates, schemas, or an external command. First-answer-wins is excluded from the primary study.
- **Cancellation:** cancellation requested, transport closure observed, and final reported usage are distinct events. Closing a stream is not assumed to stop provider billing immediately.

The named models are an initial operating point, not a claim that this ordering is universally optimal. Routing must be re-measured by task family, provider, and date.

Frozen live-risk policy

Version 0 uses a small rolling dictionary over reasoning-channel text:

- backtracking markers;
- hedging and inability-to-verify markers;
- explicit give-up markers;
- repeated four-word shingles.

The score is evaluated over the last 320 words after at least 80 reasoning words. The launch threshold is 4.0 and must be exceeded on two consecutive observations. A single word such as "maybe" cannot trigger a hedge. These are engineering defaults for offline replay, not validated production thresholds.

The policy emits its component counts, total score, words observed, and trigger time. Feature embeddings, free-form model judges, and post-hoc threshold search are excluded from v0.

Comparator arms

Tasks are assigned by generated instance or repository task, never by model call. All arms receive the same frozen context.

1. **GLM only:** wait for the primary and apply the same verifier.
2. **Always-on council:** launch GLM, Grok, and OpenAI together; wait for all configured candidates and synthesize or rank their completed answers.
3. **Fixed-delay hedge:** launch GLM, then launch both challengers after a pre-registered elapsed-time threshold if no verified result exists.
4. **Trace-triggered hedge:** launch challengers only when the frozen live-risk policy fires; otherwise allow GLM to finish alone.

If the primary completes but fails verification, both hedge arms launch their challengers regardless of whether the earlier trigger fired. This is recorded as post-completion fallback rather than early warning.

Endpoints

Correctness is a gate, not a latency trade. A hedge arm is eligible for an efficiency claim only if its task success is non-inferior to the always-on council within a pre-registered margin.

Primary endpoints:

1. p95 wall time per verified successful task;

2. billed cost per verified successful task;
3. verified task success rate.

Secondary endpoints are median latency, time from primary start to hedge, false-hedge rate, fraction of work cancelled, tokens observed before cancellation, final provider-reported usage, verifier rejection rate, council fallback rate, and winner share by task family.

The causal comparison for the trace is trace-triggered versus fixed-delay hedging at matched challenger-launch rate. Always-on council answers whether conditional fan-out preserves ensemble quality; GLM-only measures the value and cost of any fan-out.

Offline gate before spend

Replay timestamped historical traces without making API calls. For every completed primary trace, simulate when the frozen policy would have launched challengers. Proceed to a live pilot only if all of the following hold:

- at least 30 eligible tasks and 10 primary failures are present;
- the trace trigger precedes primary termination on at least 60% of failures;
- median simulated warning time is at least 15 seconds;
- false hedges occur on no more than 40% of verified primary successes;
- a matched-rate fixed-delay baseline is frozen before the live comparison.

Failure stops the live branch and is reported as **Not supported**. A separate budget and one-look stopping rule must be registered before any live council or hedge calls. No spend is authorised by this v0 document.

Safety and accounting invariants

- Candidate tasks never share a writable checkout.
- The pause hook runs once, before challenger launch.
- A winner is elected only after the same external verifier used by all arms.
- Unfinished tasks receive cancellation requests and are awaited to transport closure; absence of a final usage record is not recorded as zero cost.
- Worst-case authorised model cost, including a possible council judge, must fit below the run cap before the first request is admitted.
- Prompts and handoff context are secret-scanned before public release. Raw traces remain a separate checksummed artifact.

Reproducible interface

The released harness is `drc hedge`. A dry run performs model/config and worst-case budget admission without making calls:

```
drc hedge \
  --prompt-file task.md \
  --primary or/glm-5 \
  --challenger or/grok-4-fast \
  --challenger or/gpt-5.5 \
  --verify-command './verify-answer' \
  --cap 5 \
  --out data/hedge/run.json \
  --dry-run
```

The result JSON contains the frozen plan, risk snapshot, candidate completion records, winner, cancellation requests, and a timestamped event log. Provider usage is reported where returned; cancellation savings are never inferred from transport closure alone.

Positioning

LLM-Blender ranks and fuses completed outputs, while Mixture-of-Agents passes multiple model outputs into later aggregation layers. Those are council or ensemble designs. The systems analogue is a hedged request: start a secondary request when the primary looks slow and cancel outstanding replicas after one succeeds. This protocol replaces a fixed time-only hedge trigger with a small, auditable signal from the primary model's unfolding trajectory.